One of the most common query performance problems is inefficiently filtering down rows. For efficient filtering you'll want your database table to have an Index that matches the fields in the filter criteria.

Without those indexes, you've got a *Missing Indexes* class of performance problem. How do you find those and fix them?

Finding Missing Indexes

Missing Indexes are a type of performance problem where a query filters down a large set of rows but does that without Indexes.

Since there's no index, PostgreSQL has to scan the entire table, called a *Sequential Scan*. For tables with a row count in the millions, and when filtering just one or a few rows in your result, scanning the whole table is inefficient.

How can you discover these opportunities to add indexes and make sure they improve the query performance? One technique is to use a query like find_missing_indexes.sql[10] to analyze PostgreSQL and what it's tracked about past queries.

Find Missing Indexes shows the counts of Sequential Scans and Index Scans for a given table. When a table has a lot more Sequential Scans compared with Index Scans, the table might be missing indexes.

PgHero even has a *Suggested Indexes* capability you can use to help spot missing indexes. This tool uses row estimates for the table, and analyzes the existing table indexes and queries in order to suggest indexes to add.

Another tool that can be used is a command line tool. This tool is from the rails_best_practices[11] gem and can be used to help identify missing indexes. The gem has been added to Rideshare. From your terminal, run rails_best_practices . (the period means you're running it for the current directory) to see what it suggests. Vehicle Reservations have a Foreign Key column to Trip Requests, and Rails Best Practices suggests adding an index to that column.

Although you may wish to proactively add Indexes to tables, consider balancing that with a periodic removal of unused indexes. This process will be covered more later.

10. https://github.com/andyatkinson/pg_scripts/blob/master/find_missing_indexes.sql
11. https://github.com/flyerhzm/rails_best_practices